

Drainage and Wastewater Management Plans

Technical Summary:

Assessing Surface Water Separation

Draft DWMP (June 2022)

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Please refer to the 'Glossary of Terms' for further detail on text highlighted in green.

Purpose

The purpose of this document is to outline the methodology used to develop options for assessing surface water separation scenarios. This document is one of series of technical summaries that provide supporting detail to our stakeholders on the assessments and methodologies we have undertaken to produce the Drainage and Wastewater Management Plan (DWMP).

In this technical summary we describe the method, data sources and approach that was used in our DWMP to produce the following outputs:

- a) Identification of catchments that will provide the best opportunity for **surface water separation**
- b) Assessment of the benefits of surface water separation in reducing hydraulic capacity related sewer flooding and spills at **Storm Overflows**

Overview

Surface water runoff during rainfall events contributes vast amounts of water to the sewerage system, resulting in sewer capacity issues such as sewer flooding or spills from storm overflows to the environment. This is particularly problematic on **combined sewerage systems** which often perform as designed in dry weather but are susceptible to issues during intense rainfall. The issues surrounding surface water runoff are set to increase in the future due to the effects of climate change on rainfall intensity and the addition of new **impermeable** surfaces connecting to the system from new developments, roofs/extensions, driveways, highways and paving.

Surface water separation is an approach to disconnect impermeable area from the sewerage system, typically done by mimicking natural drainage features with **Sustainable Drainage Systems (SuDS)** such as ponds, swales and ditches. Surface water separation is considered a more robust solution compared to traditional storage tanks due to its adaptability and the wider benefits it delivers in addition to flood resilience, including water quality, amenity and **biodiversity**.

Through the **Option Development & Appraisal (ODA)** stage of the DWMP, sewerage catchments have been screened for viability around potential for surface water separation before moving to a more detailed assessment of what capacity benefits it could deliver regarding sewer flooding and Storm Overflow spills.

Description of Methodology

The methodology used to assess surface water separation potential and the benefits of reducing flooding risk and spills from Storm Overflows falls under three main stages:

- 1) Option Development and Appraisal (ODA) Screening outputs
- 2) Surface water separation modelling
- 3) Assessing the cost and benefit of surface water separation options

1) Option Development and Appraisal (ODA) Screening Outputs

To understand more about the ODA Screening process in full you can read the 'ODA Screening' Technical Summary. One of the stages of the ODA Screening was to assess what interventions are available in the catchment. This assessed a long list of potential options with the aim of providing a shortlist of feasible options. One of the available options is **Surface Water Inflow Management**, where we looked at a range of datasets to understand if there is likely to be good potential for surface water separation in a catchment. Catchments were ranked from 'Very Low' to 'Very High' opportunity. This option includes the separation of surface water inflows into the foul or combined sewer network from connections from roads and footpaths.

Surface water inflows are also common from areas of roof which drain via downpipes into the sewer network, however these can be more difficult to address on a strategic scale due to the large number of individual properties that would require disconnecting. It is therefore appropriate to start by looking at the level of opportunity for separation from roads where larger volumes of inflow may be disconnected at fewer locations.

To assess whether there is likely to be good potential for surface water separation, the ODA Screening process used a number of datasets to give an indication. The DWMP utilises the **hydraulic models** which have been built and maintained over many years. When building our hydraulic models, land and buildings within the catchment are split down into smaller **sub-catchments**. Following surveys, these areas are coded based on the type of surface such as road, footpath, or roof and whether they **infiltrate** into the ground or connect into a surface water sewer or combined sewer. Catchments with a large proportion of areas where road and footpaths connect into the combined sewer network are identified as having greater opportunity. In addition, we were able to identify using our hydraulic models, points where surface water sewers connect into the combined sewer network. Catchments with a large number of these points were identified as having good potential as these locations are likely to be quick wins when it comes to separating surface water connections.

2) Surface Water Separation Modelling

For all catchments that have a **population equivalent** greater than 2000 and where the ODA Screening intervention availability scored 'Medium', 'High' or 'Very High', additional modelling was undertaken to understand whether there was likely to be a good benefit to reducing sewer flooding in this catchment through surface water separation options. This was undertaken on 137 catchments.

The process involved replicating the **Baseline Risk and Vulnerability Assessment (BRAVA)** modelling for 10 year, 30 year and 50 year return period design storms to test three different surface water separation scenarios (**Table 1**). To understand more about the BRAVA Modelling process, see the 'BRAVA modelling' Technical Summary. For each of the three separation scenarios, a percentage of land area connecting to the combined sewer network from roads, footpaths and pavements was removed from the model run to test what impact this would have at reducing the risk of sewer flooding in the catchment. This was recorded in hectares. The model was run using the 2050 future model which includes future pressures such as climate change, growth, and urban creep. It gave the outputs as a volume of flooding predicted from manholes in the catchment. For all surface water separation scenarios, the model was run assuming a 15% uplift in rainfall due to climate change.

Table 1. Summary of the surface water separation scenarios

Scenario	Explanation
Realistic – 10% reduction in surface water runoff area	This represents what is likely to be deliverable based on current ways of working
Stretch – 30% reduction in surface water runoff area	This represents an indication of what could be achieved if organisations work together to better deliver surface water management strategies
Aspirational – 50% reduction in surface water runoff area	This represents what could be achieved with changes to legislation to make it easier for a sewerage company to build Sustainable Drainage Systems (SuDS) as part of a wider surface water management remit

The surface water separation modelling provides a volume of flooding from manholes across the catchment which could be compared against the 2050 BRAVA model flood volumes to give an understanding of the total flood volume reduction that could be achieved by separating surface water inflows for the different scenarios. This helped us to identify which catchments have the best opportunity to reduce sewer flooding risk by separating surface water. **Figure 1** summarises this process.

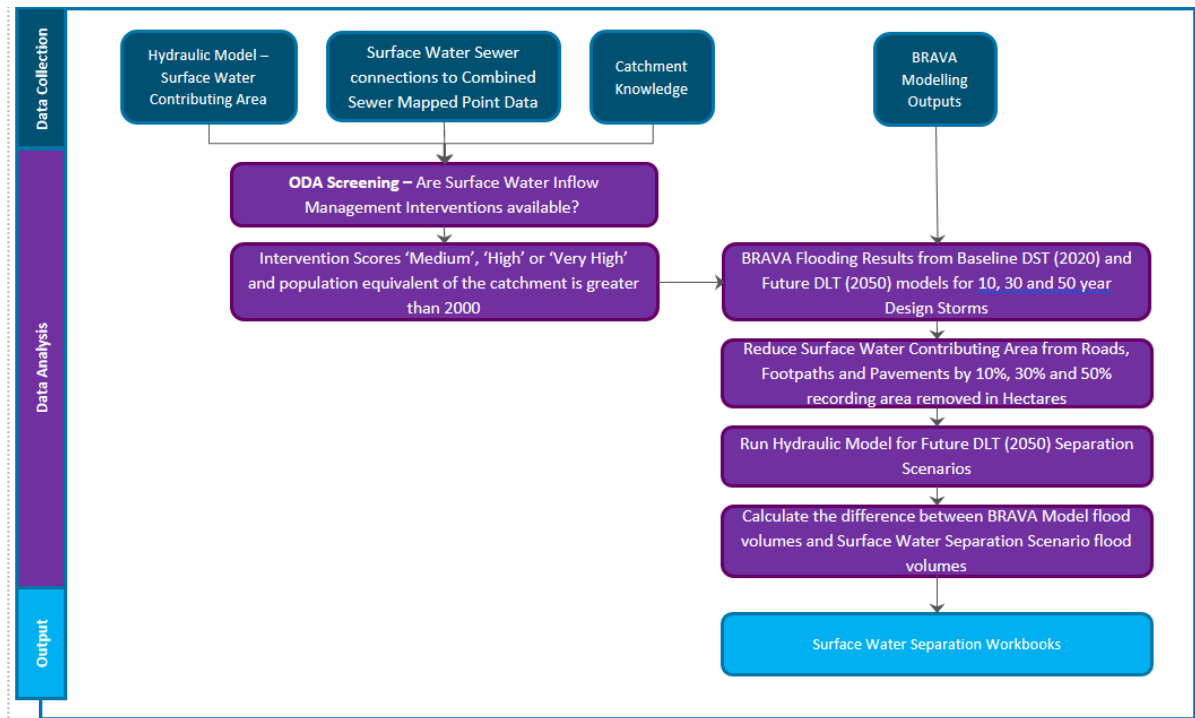


Figure 1. Surface water separation process flow chart for steps 1 and 2

3) Assessing the cost and benefit of Surface Water Separation Options

Taking the total area (hectares) of impermeable area removed from each separation option (10%/30%/50%) a basic cost estimate was produced by multiplying it with a unit cost per hectare.

It is recognised that surface water separation alone will often not be sufficient to fully alleviate sewer flooding. In these instances, it is assumed the separation will be supplemented with traditional sewer capacity upgrades, recognising this is one of the most prevalent interventions to come out of the ODA Screening exercise. This combination of separation and traditional type interventions is referred to as a hybrid of green Sustainable Drainage Systems (SUDS) and grey (traditional, often concrete) solutions. Hybrid solutions can be used to alleviate sewer flooding and a method for cost estimating the grey/traditional element of them can be seen in the 'Internal Sewer Flooding' Technical Summary.

Outputs

The surface water separation assessments fall within the ODA stage of the DWMP with the main purpose to identify which catchments have the most potential to address future flood risk through the removal of surface water entering the combined sewer network. The ODA Screening helped us to narrow down our assessments to the catchments with the most potential.

The outputs from the ODA surface water separation modelling exercise demonstrated that for most catchments a reduction in future flood risk can be obtained by separating surface water connections from the combined sewer network. The level of benefit that can be achieved varies between catchments based on catchment characteristics and the amount of area in hectares that is removed.

What this tells us

The assessment of surface water separation opportunity as part of the ODA process demonstrates that there is opportunity for a good number of catchments to reduce their future sewer flood risk by separating surface

water inflows that currently connect into the combined sewer network. The level of opportunity varies between catchments, however in general the greater the area in hectares of separation the greater the reduction in flood volumes from manholes. Catchments identified as having the most potential to reduce sewer flooding should therefore be taken forward as catchment strategies, working alongside other [Risk Management Authorities](#) such as Local Councils, Highways Authority and [Lead Local Flood Authorities](#) to explore options to identify specific locations where separation and SuDS may be viable.

The management of surface water inflows is deemed to be a sustainable approach to managing future flood risk from sewers as it can have multiple additional benefits. These include the environmental, social and economic benefits of reduced hydraulic sewer flood risk, the reduction in spills from storm overflows to the environment, water quality benefits, amenity and biodiversity benefits to local communities and a reduction in flows to Wastewater Treatment Works that require treating. Therefore, catchment strategies that can focus on the separation of surface water should be pursued in more detail.